`Hello WORLD`. Finally, token filters are applied so that the sentence can be transformed into tokens: `[hello] [world]`. By default, Elasticsearch uses a standard analyzer to lowercase the tokens, as is the case here. Note that the punctuation (comma) was also removed during this process.

After these steps, an inverted index is created for this field. It is predominantly used in full-text search functionalities. In essence, it is a hash map with words as keys pointing to documents where these words are present. It consists of a set of unique words and the frequency with which those words occur across all documents in the index.

Let's revisit our example. Because document 1 ("Hello WORLD") was indexed and analyzed, an inverted index was created with the tokens (individual words) and the documents they occur in; see table 3.3.

Table 3.3 Tokenized words for the "Hello, World" document

Word	Frequency	Document ID
hello	1	1
world	1	1

The words *hello* and *world* are added to the inverted index along with the document IDs where these words are found (of course, document ID 1). We also note the frequency of the words across all documents in this inverted index.

When the second document ("Hello, Mate") is indexed, the data structure is updated (table 3.4).

Table 3.4 Tokenized words for the "Hello, Mate" document

Word	Frequency	Document ID
hello	2	1,2
world	1	1
mate	1	2

When updating the inverted index for the word *hello*, the document ID of the second document is appended, and the frequency of the word increases. All tokens from the incoming document are matched with keys in the inverted index before being appended to the data structure (like *mate* in this example) as a new record if the token is seen for the first time.

Now that the inverted index has been created from these documents, when a search for *hello* comes along, Elasticsearch will consult this inverted index first. The inverted index points out that the word *hello* is present in document IDs 1 and 2, so the relevant documents will be fetched and returned to the client.